

YUWEI CHENG

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EDUCATION

PhD	University of Chicago Department of Statistics	GPA 3.8/4.0, 2021-2026, Jun
	ICML26 Position Track Silver Reviewer	
	NeurIPS Scholar Award 2025	
	Best Poster Award at Midwest Machine Learning Symposium 2024	
	Spring Quarter Consulting Cup Winner (Best team in the Statistical Consulting Program)	
BS	National University of Singapore Department of Statistics	GPA 4.9/5.0, 2016-2020
BA	National University of Singapore Department of Economics	GPA 4.9/5.0, 2016-2020
	Dean's List Recipient (GPA ranked in the top 5% for 4 consecutive years in both departments)	
	Science and Technology Scholarship (Full scholarship covering tuition and living expenses for 4 years)	

RESEARCH INTEREST

Preference Alignment, Reinforcement Learning, Large Language Model, Post-Training, Statistical Modeling

INDUSTRIAL EXPERIENCE

Data Scientist Research Intern at Trust and Safety Team of **YouTube, Google** Jun 16, 2025 – Sep 19, 2025

- Researched and designed **evaluation dimensions and metrics** for AI-generated metric report summaries, contributing to the automation of the metric reporting process.
- Developed a **Python library** implementing the **proposed LLM-based rater with prompting strategies that evoke chain-of-thought reasoning** for improved interpretability and diagnostic insights.
- Conducted comprehensive sensitivity analyses to evaluate metric fidelity and collected human-written summary samples to align automated ratings with human preferences.
- Generated scorecards for AI-produced summaries and provided product improvement recommendations based on the evaluation results.

PUBLICATIONS

*indicates co-first author

[W2] **Cheng, Y*, Tian, W*, & Xu, H. (2026+). Fine-Tuning Improves Information Conveyance in Language Models.** Under Review.

[W1] Wang, Q., **Cheng, Y.**, Song, A., Keedy, S., Berman, M., & Feamster, N. (2025+). **Can LLMs Address Mental Health Questions? A Comparison with Human Therapists.** Under Review.

[P7] Wang, G, **Cheng, Y.**, & Xu, H. (2025+). **Exponential Convergence of Probabilistic Bisection Algorithm with Noisy Labels**. ICML 2026: Forty-Third International Conference on Machine Learning, 2026.

[P6] Yi, B., Liu, Q., **Cheng, Y.**, & Xu, H. (2026). **Escaping Model Collapse via Synthetic Data Verification: Near-term Improvements and Long-term Convergence.** ICLR 2026: Proc. 14th International Conference on Learning Representations, 2026.

[P5] **Cheng, Y., Zhao, Z., & Xu, H. (2025). Learning Personalized Ad Impact via Contextual Reinforcement Learning under Delayed Rewards.** NeurIPS 2025: The Thirty-Ninth Annual Conference on Neural Information Processing Systems

[P4] Yao, F*, **Cheng, Y.***, Wei, Er., & Xu, H. (2025). **Single-Agent Poisoning Attacks Suffice to Ruin Multi-Agent Learning**. ICLR 2025: Proc. 13th International Conference on Learning Representations, 2025.

[P3] **Cheng, Y.**, Yao, F., Liu, X., & Xu, H. (2025). **Learning from Imperfect Human Feedback: a Tale from Corruption-Robust Dueling**. ICLR 2025: Proc. 13th International Conference on Learning Representations, 2025.

[P2] Quaye, S. E. D., **Cheng, Y.**, Tan, R. K. J., Koo, J. R., Prem, K., Teo, A. K. J., & Cook, A. R. (2023). **Application of the network scale-up method to estimate the sizes of key populations for HIV in Singapore using online surveys**. *Journal of the International AIDS Society*, 26(3), e25973.

[P1] **Cheng, Y.**, et al. (2022). **Estimates of Japanese encephalitis mortality and morbidity: a systematic review and modeling analysis**. *PLOS Neglected Tropical Diseases*, 16(5), e0010361.

SKILLS & SERVICES

Programming: Python, shell scripting, R

Data Analysis: Generalized linear models, Gradient boosting, SVM, Principal component analysis, Neural network, Sentiment analysis, Causal inference, Conformal prediction, Experimentation

Teaching Experience:

- Instructor for Elementary Statistics with 45+ students Winter 2025
- Teaching assistant for Applied Regression Analysis Fall 2024
- Teaching assistant for Introduction to Data Science Fall, Winter 2021-2024

Professional Services:

- Program Committee Members of AI for Mechanism Design and Strategic Decision Making (AIMS) workshop at ICLR26 2026
- Reviewer for ICML, NeurIPS, ICLR 2025, 2026
- Reviewer for The American Statistician 2024
- Reviewer for BMJ Global Health 2023

TALKS

Towards Development and Assessment of Evaluation Metrics for AI Generated Metric Trend Summary

- Google PhD Summit Poster Session (July, 2025)

Learning Personalized Ad Impact via Contextual Reinforcement Learning under Delayed Rewards

- Salesforce AI Research Future Forum (May, 2025)
- University of Chicago, research talk at Department of Computer Science Theory Lunch (May, 2025)

Intrinsic Efficiency-Robustness Trade-offs in Modern Online Learning Algorithms

- National University of Singapore, Department of Computer Science (April, 2025)
- Nanyang Technological University, Agent Mediated Intelligence Research Group (April, 2025)